

Bicomplex Signal Geometry: Completed Source Charts, Monodromy and Spectral Stability

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Abstract

We study an explicit ruled map associated with a two-component signal model. A completed source atlas includes its two previously omitted endpoint singularities. The planar observation field has two interior zeros of opposite index; continuous square-root lifting gives a sign change on one traversal and restoration on two. The standard fold and cross-cap residuals have sublevel exponent $3/4$. Their real links, an explicitly chosen complex A_3 completion, and the actual principal-curvature cusp are treated as distinct constructions. On a fixed Lebesgue Hilbert space we prove the Mellin integration-by-parts identity, identify the half-line domain obstruction, and establish finite-time Gram estimates. Distinct finite frequencies are independent for every positive observation time. We also prove local zero capacity and explain why it is a statement about closed forms rather than minimal operator domains. Weighted radial estimates give compactness and a positive fixed-Dirichlet gap on the completed source, including its endpoint corners. An audit of the later operator claims exposes a concrete domain error: closing the interior compact core leaves infinitely many outer-boundary deficiency modes. We give a corrected conditional construction from a Dirichlet realization and bounded point traces. The actual Green and twisted Dirac parametrices require further estimates, which are stated explicitly. The normalization defect, the phase sign line, and a chosen full-spin transmission map retain different data and roles. No communication capacity, particle assignment, or arithmetic weight follows from a numerical analogy.

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1 The model and its mathematical inputs

The geometric object in this paper is the map

$$S(t, u) = (c + 2u(1 - c), (1 - u)s, uQ), \quad c = \cos t, \quad s = \sin t, \quad Q = \sqrt{c(2 - c)}, \quad (1)$$

with physical labels $D = [-\pi/2, \pi/2] \times [0, 1]$. We keep source labels distinct even when their images coincide. In particular, $S(t, 1) = S(-t, 1)$ is an image relation, not a quotient defining the source. The term bicomplex can be made precise by taking commuting units i, j with $i^2 = j^2 = -1$ and idempotents $e_{\pm} = (1 \pm ij)/2$. The algebra then identifies with $\mathbb{C}e_+ \oplus \mathbb{C}e_-$. Choosing the coefficient norm $|z_+|^2 + |z_-|^2$ is additional Hilbert data; it is not a multiplicative bicomplex modulus. None of the proofs below depends on bicomplex multiplication.

The momentum map $(z_+, z_-) \mapsto (|z_+|^2, |z_-|^2)$ and the unit-power constraint define a sphere and its moment interval. They do not, by themselves, define a two-dimensional source, a lift of (1), or a projection taking that lift to S . We therefore take (1) as an explicit model input. A signal realization through an ambient projection would have to supply those maps separately. This resolves a missing definition in historical v12, rather than declaring that an unspecified map exists.

We use conjugate-linear first inner products throughout: $\langle f, g \rangle = \int \bar{f}g$. A fold in this paper means a smooth plane-to-plane germ equivalent to $(x, y) \mapsto (x, y^2)$. A cross-cap means a smooth plane-to-three-space germ equivalent to $(x, y) \mapsto (x, xy, y^2)$. These dimension-specific normal forms are not definitions for arbitrary maps $\mathbb{R}^n \rightarrow \mathbb{R}^m$.

2 The completed source and its five singular labels

The angular formula is smooth where $c > 0$ but fails to be a smooth chart at $Q = 0$. This defect is repaired by changing the source coordinate, while keeping the full physical surface [1].

Proposition 2.1 (Endpoint charts and complete rank set).

Near either endpoint let $\epsilon \in \{1, -1\}$, $r = Q \geq 0$, and set

$$C(r) = 1 - \sqrt{1 - r^2}, \quad B_{\epsilon}(r, u) = (C + 2u(1 - C), \epsilon(1 - u)\sqrt{1 - C^2}, ur).$$

The formula extends smoothly to $|r| < 1$ near $r = 0$, agrees with S on the physical overlap, and satisfies

$$(B_{\epsilon})_r(0, u) = (0, 0, u), \quad (B_{\epsilon})_u(0, u) = (2, -\epsilon, 0), \quad (B_{\epsilon})_r \times (B_{\epsilon})_u = (\epsilon u, 2u, 0).$$

The rank-loss labels of the completed source are exactly

$$P_0 = (0, 1), \quad P_{\pm} = (\pm t_0, u_0), \quad E_{\pm} = (\pm \pi/2, 0), \quad c_0 = \frac{3 - \sqrt{5}}{2}, \quad t_0 = \arccos c_0, \quad u_0 = \frac{1 - c_0}{2}.$$

Their smooth extensions are cross-caps. The physical restrictions at P_0 and $E_{\pm}$ are respectively a boundary half-germ and corner quarter-germs.

Proof. The radicands are positive near $r = 0$. Since $C(r)(2 - C(r)) = r^2$, $C(Q) = c$ when $0 \leq c \leq 1$; the sign of s fixes ϵ . Differentiation gives the displayed endpoint vectors, so an endpoint loses rank exactly at $u = 0$. On $c > 0$, the full cross-product equations reduce either to $c = 1, u = 1$, or to $u = (1 - c)/2 = c(2 - c)/2$. The latter equation is $P(c) = c^2 - 3c + 1 = 0$ and has exactly one root in $(0, 1)$. Direct substitution checks the remaining component. Thus $P(c)$ selects the lateral points, not all surface singularities.

For a rank-one ruled germ choose the nonzero ruling vector and its kernel direction. Because $S_{uu} = 0$, the cross-cap recognition determinant reduces to $\det(S_u, S_{ut}, S_{tt})$. Its values are

$$\Delta(P_0) = 1, \quad \Delta(P_{\pm}) = -5\sqrt{\sqrt{5} - 2} \neq 0.$$

In the endpoint chart, $\det(B_u, B_{ur}, B_{rr})(0, 0) = -\epsilon \neq 0$. The smooth cross-cap criterion applies to these actual maps with their adapted directions [2, Corollary 4.5 and its proof]. It supplies smooth source and target diffeomorphisms, beyond a two-jet resemblance. The half- and quarter-germs refer to the original physical inequalities in those charts; no coordinate straightening of their boundaries is assumed. $\square$

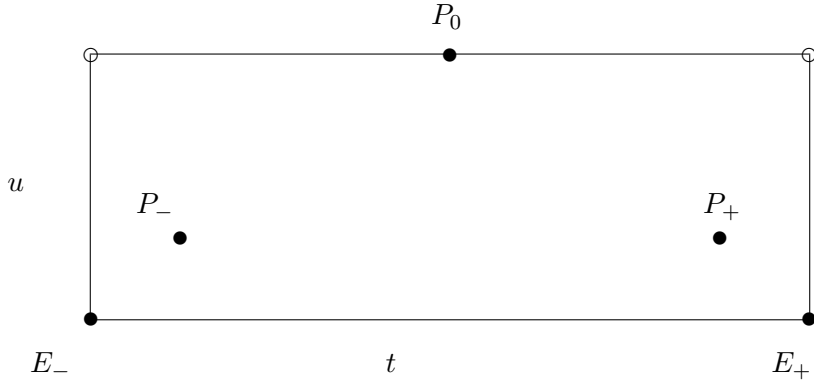


Figure 1: Filled markers are the five rank-loss source labels; open markers are the two regular upper corners, which remain distinct source labels. Positions are drawn from t_0, u_0 of Proposition 2.1; this parameter diagram is not a surface embedding.

Put $D_{\text{reg}} = D \setminus \{P_0, P_+, P_-, E_+, E_-\}$ with the completed atlas. It is a connected orientable surface with retained regular boundary corners. Deleting boundary points from a disk does not create an interior hole: one may retract collar arcs into the interior. Removing the two interior points leaves a twice-punctured disk. For example, excise two disjoint small disks and retract the remaining disk onto two circles joined to a base point by disjoint arcs. Hence $H_1(D_{\text{reg}}; \mathbb{Z}) = \mathbb{Z}^2$, and its orientation cover is the disconnected trivial cover. Orientation comes from the source charts; there is no need for an invalid angular normal at $Q = 0$. The paired top-edge image relation has a fixed point and supplies no free boundary gluing.

3 The actual observation field and square-root lifting

On the open angular chart $|t| < \pi/2$ define

$$N = (1 - c)(1 - c - 2u), \quad M = \frac{c^2(2 - c) - u(1 - c + c^2)}{Q}, \quad F = N + iM. \quad (2)$$

Both N and M are even in t . Their Jacobian determinant is odd, because the t derivatives reverse sign. A source area reparametrization multiplies this field by its Jacobian. The angular field generally diverges at $Q = 0$; it is not a continuous global scalar field on the completed rectangle.

Theorem 3.1 (Interior dipole and restoration).

The interior zeros of F are exactly $P_{\pm}$. With the (t, u) orientation,

$$\det DF(P_{\pm}) = \mp \frac{5s_0(1 - c_0)^2}{Q_0}, \quad \text{Ind}(P_{\pm}) = \mp 1.$$

Every continuous square-root branch transported round a sufficiently small positive loop about either point changes sign. Two traversals restore it.

Proof. Solving (2) gives the same lateral roots as above. The determinant formula follows by substitution and $c_0^2 = 3c_0 - 1$. It is nonzero. The actual small-circle winding of a smooth planar nondegenerate zero is the sign of its determinant: a linear homotopy to the derivative avoids zero on sufficiently small circles; an invertible real linear map has winding 1 or -1 according to orientation. The predecessor proof applies to exactly (2), the same open domain and orientation [1].

Along a loop write $F(\gamma(s)) = \exp L(s)$ using a continuous logarithm lift. Its endpoint is $L(1) = L(0) + 2\pi i n$, where $n = \pm 1$. Thus $\chi(s) = \exp(L(s)/2)$ satisfies $\chi(1) = (-1)^n \chi(0) = -\chi(0)$. Any other continuous root divided by χ is a continuous map from a connected interval to $\{1, -1\}$ and is constant. Repeating the loop makes the sign positive. This is a traversal count; it does not assign a physical clock or equate the root cover with the normalization-sheet cover. $\square$

Proposition 3.2 (A genuine polar fold).

The smooth extension of $\Psi = (N, M)$ at P_0 is a fold. It has full-extension local degree zero. No interior winding index is assigned on its physical half-neighbourhood.

Proof. Use source coordinates

$$a = -M(t, u), \quad b = 2 \sin(t/2) \sqrt{u - (1 - \cos t)/2},$$

and target coordinates $(N, M) \mapsto (-M, -N)$. Near $(0, 1)$ the source radicand is positive, and its derivative matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is invertible. The inverse-function theorem gives two-sided smooth charts. The exact identity $-N = b^2$ puts the map into (a, b^2) ; thus no conclusion is inferred from a Taylor jet alone. Reflection of b pairs regular preimages with opposite signs, or, equivalently, the model circle lies in a half-plane and has zero winding. The target/source orientation changes do not change zero degree. In historical v12 the displayed remainder for M omitted $M(t, 1) = -t^4/2 + O(t^6)$; this term does not invalidate these charts. $\square$

4 Residual volume and an explicit complex completion

For the standard models take source Lebesgue measure and residuals $K_f = x^2 + y^4$ and $K_{\times} = x^2(1 + y^2) + y^4$.

Theorem 4.1 (Sublevel law with quantitative comparison).

Writing $B = B(1/4, 3/2)$, one has, for $\varepsilon > 0$,

$$V_f(\varepsilon) = B\varepsilon^{3/4}, \quad \frac{B\varepsilon^{3/4}}{\sqrt{1 + \sqrt{\varepsilon}}} \leq V_{\times}(\varepsilon) \leq B\varepsilon^{3/4}.$$

The common exponent is $3/4$. A smooth positive density changes the leading constant by its value at the origin. Smooth invertible source and target changes preserve the exponent, but not generally the displayed coefficient.

Proof. Slicing in y gives $4 \int_0^{\varepsilon^{1/4}} \sqrt{\varepsilon - y^4} dy$ for the fold. Set $y = \varepsilon^{1/4}v$ and then $z = v^4$ to obtain $B\varepsilon^{3/4}$. For the cross-cap the integrand has the additional factor $(1 + y^2)^{-1/2}$; on this interval it lies between $(1 + \sqrt{\varepsilon})^{-1/2}$ and 1. After anisotropic scaling the density converges uniformly to its value at the origin. A target diffeomorphism fixing zero compares squared norms by positive constants; a source diffeomorphism has a bounded positive Jacobian. These comparisons retain the power exponent. They do not make the standard Euclidean coefficient an invariant of arbitrary coordinates or metrics. $\square$

The corresponding geometric cost satisfies $-\log V(\varepsilon) = \frac{3}{4} \log(\varepsilon^{-1}) - \log B + o(1)$ in the specified standard measure. Separately, $|P'(c_0)| = \sqrt{5}$ for the monic $P(c) = c^2 - 3c + 1$; multiplying P by a nonzero constant changes that derivative.

Theorem 4.2 (Chosen A_3 completion and its cohomology).

Extend the coefficients of $K_\times$ to $\mathbb{C}$. Its holomorphic germ is right-equivalent to $X^2 + Y^4$, with Milnor and Tjurina numbers three. The affine fiber $U = \{X^2 + Y^4 = 1\}$ compactifies to a smooth genus-one curve E with two points removed. Its first cohomology has dimension three and monodromy spectrum $\{-i, 1, i\}$. Its mixed Hodge weight-graded dimensions in weights one and two are respectively two and one.

Proof. On a disk $|y| < 1$ choose the holomorphic square root of $1 + y^2$ taking value one at zero. The coordinates $X = x\sqrt{1 + y^2}, Y = y$ have invertible derivative and give the equality exactly. The Jacobian algebra is $\mathbb{C}\{Y\}/(Y^3)$; weighted Euler's identity puts $X^2 + Y^4$ in its Jacobian ideal, giving the same Tjurina dimension.

The weighted projective compactification is $X^2 + Y^4 = Z^4$ in $\mathbb{P}(2, 1, 1)$. It avoids the ambient weighted singular point. The map to $[Y : Z] \in \mathbb{P}^1$ is a double cover branched at four simple roots. Riemann–Hurwitz gives genus one. At infinity $W = X/Y^2$ takes the two distinct values $\pm i$, so both ends are smooth and unramified. The first Betti number is $2g + 2 - 1 = 3$. Weighted homogeneity identifies this fiber with the local Milnor fiber [3, 4]. The monodromy $h(X, Y) = (-X, iY)$ pulls dY/X back to $-i dY/X$ and acts by i on its conjugate. Near infinity $(s, W) = (1/Y, X/Y^2)$ transforms as $(-is, W)$, so the boundary class is fixed. Finally the Gysin sequence of mixed Hodge structures is

$$0 \longrightarrow H^1(E; \mathbb{Q}) \longrightarrow H^1(U; \mathbb{Q}) \longrightarrow \tilde{H}^0(\{\infty_+, \infty_-\}; \mathbb{Q})(-1) \longrightarrow 0.$$

The first term is pure of weight one with types $(1, 0), (0, 1)$; the last has type $(1, 1)$ and weight two [5, 6]. These are external cohomological theorems with their required smooth algebraic pair explicitly constructed. No Hodge structure is inferred from the real surface or a Mellin parameter. $\square$

5 Three different projected curves

Proposition 5.1 (Real link and embedded lift).

For every $r > 0$, $L_r = \{K_\times = r^2\}$ is a smooth circle. Its cross-cap image has one transverse double point and is homeomorphic to $S^1 \vee S^1$. The lift (x, xy, y^2, y) is a smooth embedding of the source plane.

Proof. The gradient of $K_\times$ vanishes only at zero. The compact level set is the two graphs $x = \pm\sqrt{(r^2 - y^4)/(1 + y^2)}$, joined at $y = \pm\sqrt{r}$, and is connected. A connected compact smooth one-manifold is a circle. Equality of cross-cap images forces $x = x'$ and $y' = \pm y$; distinct preimages on the link occur only at $(0, \pm\sqrt{r})$. The tangent images there are $(1, \pm\sqrt{r}, 0)$, which are transverse. Elsewhere the source map has rank two, so the link restriction is immersed. The compact-domain quotient identifies exactly that pair and gives the figure-eight graph. The first and fourth lift coordinates supply a continuous smooth inverse, and their differential has rank two everywhere. This proves an embedding, rather than injectivity alone. $\square$

Proposition 5.2 (Weighted orbit and its amplitude domain).

The primitive positive weights of $X^2 + Y^4$ are $(2, 1)$. For $a, b > 0$, $a^2 + b^4 = 1$, the orbit $\Gamma(s) = (ae^{2is}, be^{is})$, $s \in \mathbb{R}/2\pi\mathbb{Z}$, is embedded. Its rectangular shadow $(b \sin s, a \sin 2s)$ is a transverse figure-eight; adding $\Re Y$ reconstructs the orbit. The frequency ratio is $2 : 1$. The quadratic phase action is $2\pi(4a^2 + b^2)$. Its infimum on nondegenerate orbits is 2π , attained only in their degenerate closure; its unique interior critical point $b^2 = 1/8$ is a strict maximum.

Proof. The equation $2w_X = 4w_Y$ and coprimality give $(2, 1)$. The Y coordinate determines the phase, so the lifted circle is embedded. The shadow has only the distinct pair $s = 0, \pi$ over its double point; the tangent determinant is $4ab \neq 0$. Eliminating phase gives $v^2 = 4a^2(u^2/b^2)(1 - u^2/b^2)$, and adding $b \cos s$ fixes phase. Differentiation gives $X'' + 4X = 0$, $Y'' + Y = 0$. With $z = b^2 \in (0, 1)$ the action divided by 2π is $4 + z - 4z^2 = 1 + (1 - z)(4z + 3) > 1$. Its derivative vanishes only at $z = 1/8$ and its second derivative is -8 . The value 1 occurs at $z = 1$ in the closed family. Replacing s by $\omega_0 t$ changes the clock for every $\omega_0 > 0$. The orbit leaves the fixed fiber $f = 1$ at general phases: $f(\Gamma(s)) = e^{4is}$. It is an orbit in the ambient completion. $\square$

Proposition 5.3 (Actual principal-curvature cusp).

For the actual lateral germs, the Euclidean Whitney invariants obey $b_\pm^2 + a_\pm c_\pm^2 = 0$ with $c_\pm > 0$. Their principal-curvature exceptional traces have an ordinary cusp at the double return, rather than a transverse node.

Proof. Use the actual adapted pair $\xi = \partial_u/\sqrt{3}$, $\eta = -\partial_t + \lambda_\pm \partial_u$, $\lambda_\pm = -c_0 s_\pm/[2(1 - c_0)]$. Let Π project onto the normal plane of S_u , and put $A = \Pi(S_{tt} - 2\lambda_\pm S_{tu})$, $B = \Pi(-S_{tu}/\sqrt{3})$. The independent derivative computation yields

$$\begin{aligned} N &= |A|^2 = \frac{-335 + 162\sqrt{5}}{12}, \\ M^2 &= \langle A, B \rangle^2 = \frac{-520 + 249\sqrt{5}}{27}, \\ R &= |B|^2 - M^2/N = \frac{20(140 + 11\sqrt{5})}{11397} > 0. \end{aligned}$$

Oriented Gram–Schmidt gives $a = -M^2/(2N^{3/2})$, $b_\pm = \pm\sqrt{M^2 R}/N$, $c = \sqrt{2R}/N^{1/4}$. Hence the discriminant vanishes exactly. The external Euclidean normal-form and pedal identities [7, Proposition 2.1 and Eq. (2.8)] apply to these nondegenerate cross-caps. In their notation the pedal trace is

$$p(\tau) = \frac{2(c\tau^2 + 2b\tau - ac)}{c^2 + 4\tau^2}(-c, 2\tau).$$

Set $\tau_* = -b/c$. Its numerator is $c(\tau - \tau_*)^2$. Writing $p(\tau_* + h) = h^2 v(h)$, one has $v(0) \neq 0$ and

$$\det(p''(\tau_*), p'''(\tau_*)) = -\frac{96c^3}{(c^2 + 4b^2/c^2)^2} \neq 0.$$

This proves the ordinary cusp criterion, beyond the vanishing discriminant. It concerns the exceptional trace specified by the pedal theorem; it is not a global classification of the whole principal-curvature surface. $\square$

Construction	Source	Projected feature
Real cross-cap link	$K_\times = r^2$	transverse node
Weighted phase orbit	circle in $\mathbb{C}^2$	transverse node
Actual exceptional trace	adapted direction τ	ordinary cusp

Table 1: Distinct maps produce distinct curves. No amplitude, clock, or scattering rule transfers between these constructions.

6 Mellin integration, measures and operator domains

Theorem 6.1 (Formal Green identity and measure dependence).

Let $T_\sigma f = -i(xf' + \sigma f)$, $\sigma \in \mathbb{R}$. For complex C^1 functions on $[a, b] \subset (0, \infty)$,

$$\int_a^b (\overline{T_\sigma f} g - \overline{f} T_{1-\sigma} g) dx = i[x\overline{f}g]_a^b.$$

Thus on $C_c^\infty((1, \infty)) \subset L^2(dx)$ the formal adjoint expression is $T_{1-\sigma}$ and symmetry holds exactly at $\sigma = 1/2$. For measure $x^{2\beta-1} dx$, the formal adjoint parameter is $2\beta - \sigma$, so the symmetry parameter is $\sigma = \beta$.

Proof. The first integrand is $i(x\overline{f}g)'$; integrating gives the endpoint term. Compact support permits an interval with zero endpoint terms and then the actual improper integral equality. The compact core is dense by local truncation and smooth approximation in L^2 . On a weighted space, $(xx^{2\beta-1})'/x^{2\beta-1} = 2\beta$ replaces the constant one in the same integration by parts. Necessity of the formal coefficient equality can be tested on a nonzero compactly supported $f = g$. The weighted inner product in the historical introduction therefore cannot be silently used for the Lebesgue threshold: if $\beta = \sigma$, the weighted expression is formally symmetric for every σ . $\square$

Proposition 6.2 (Actual half-line closures and adjoints).

For $\sigma = 1/2$ define $(Uf)(s) = e^{s/2}f(e^s)$ from $L^2([1, \infty), dx)$ to $L^2([0, \infty), ds)$. It is unitary and takes the compact-core operator to $-i\partial_s$. Its closure has domain $H_0^1(0, \infty)$, its adjoint has domain $H^1(0, \infty)$, and its deficiency indices are $(1, 0)$. On the full multiplicative line the corresponding closed domain is carried to $H^1(\mathbb{R})$; that operator is self-adjoint.

Proof. The substitution $x = e^s$ proves norm equality and supplies the inverse; the chain rule gives the operator identity. The graph norm of $-i\partial_s$ is exactly the H^1 norm, so closure of the compact core is H_0^1 . Distributional integration by parts characterizes the adjoint domain as H^1 with no endpoint constraint. The adjoint Green term is $-i\overline{h(0)}k(0)$ under our convention; there is no term at infinity for H^1 functions. The equations $-ih' = \pm ih$ give $h = e^{-s}$ and $h = e^s$; only the first is square-integrable. Unequal indices exclude every self-adjoint extension. On $\mathbb{R}$ neither solution is square-integrable, and the Fourier transform identifies the closure with multiplication by real frequency on its maximal domain [8].

Alternatively the Haar-to-Lebesgue map $Wf(x) = x^{-1/2}f(x)$ is unitary from $L^2(dx/x)$ to $L^2(dx)$ on the full group and conjugates $-ix\partial_x$ to $T_{1/2}$. Restriction to a half-line does not preserve the full-line self-adjointness result. $\square$

Direct differentiation gives $T_\sigma x^{-\sigma-i\omega} = -\omega x^{-\sigma-i\omega}$. Its half-line norm squared is $\int_1^\infty x^{-2\sigma} dx$, finite exactly for $\sigma > 1/2$. Even then its trace at $x = 1$ is one, so it is outside the minimal zero-trace domain. At $\sigma = 1/2$ it is a generalized eigenfunction, not an L^2 vector. Real formal eigenvalues alone imply no orthogonality.

7 Actual finite-time Gram integrals

Let $\omega_1, \dots, \omega_M$ be distinct real frequencies, $T > 0$, and $\psi_j(t) = e^{-i\omega_j t}$. Set

$$G_{jk} = T^{-1} \int_0^T \overline{\psi_j(t)} \psi_k(t) dt.$$

This convention makes the quadratic form $c^* G c$ the norm squared of the synthesis $\sum_j c_j \psi_j$, divided by T .

Theorem 7.1 (Independence and quantitative spectral enclosure).

For $M \geq 2$ put $\delta = \min_{j \neq k} |\omega_j - \omega_k| > 0$ and $\varepsilon = 2(M-1)/(T\delta)$. Then

$$G_{jj} = 1, \quad G_{jk} = \frac{e^{iT(\omega_j - \omega_k)} - 1}{iT(\omega_j - \omega_k)}, \quad |G_{jk}| \leq \frac{2}{T|\omega_j - \omega_k|} \quad (j \neq k), \quad \|G - I\|_{2 \rightarrow 2} \leq \varepsilon.$$

The phasors are independent and G is positive definite for every $T > 0$. If $\varepsilon < 1$, then

$$1 - \varepsilon \leq \lambda_j(G) \leq 1 + \varepsilon, \quad \kappa_2(G) \leq \frac{1 + \varepsilon}{1 - \varepsilon}, \quad M \log(1 - \varepsilon) \leq \log \det G \leq 0.$$

In particular $-\log \det G \leq M\varepsilon/(1 - \varepsilon)$. For $M = 1$, $G = (1)$; an empty family has empty Gram matrix and determinant one, with no condition number or separation minimum required.

Proof. The entries are actual elementary integrals. The numerator has modulus at most two. The matrix is Hermitian. Each row of $G - I$ has absolute sum at most ε , so Gershgorin places all its eigenvalues in $[-\varepsilon, \varepsilon]$. The spectral theorem for a finite Hermitian matrix then gives the Euclidean operator norm, not just an arbitrary matrix norm.

For independence, a synthesis vanishing almost everywhere on $[0, T]$ vanishes everywhere by continuity. Differentiate at any interior time t_* to orders $0, \dots, M-1$. The coefficient vector $(c_j e^{-i\omega_j t_*})_j$ is killed by the Vandermonde matrix of the distinct numbers $-i\omega_j$. Its determinant is nonzero, so $c = 0$. This proves positivity for all positive T ; the separation threshold is only a sufficient conditioning estimate. Multiplying the eigenvalue lower bounds gives the log-determinant lower bound. Hadamard's inequality applied to the positive Gram matrix with unit diagonal gives $\det G \leq 1$. Finally $-\log(1 - \varepsilon) \leq \varepsilon/(1 - \varepsilon)$ follows by integrating $(1 - s)^{-1}$ on $[0, \varepsilon]$. $\square$

For distinct positive integers take $\omega_j = \log n_j$. Multiplicative independence is unnecessary. These estimates compare a fixed finite family as $T \rightarrow \infty$; they are not a uniform theorem for growing frequency sets.

8 Hardy evaluation and Bohr–Haar completion

Define the Dirichlet Hilbert space by its coefficient norm: $\mathcal{H}_D^2 = \{(a_n) : \sum |a_n|^2 < \infty\}$, with Dirichlet evaluation initially defined on finite polynomials. For $\Re s = \sigma > 1/2$, Cauchy–Schwarz gives $|\sum a_n n^{-s}| \leq \|a\|_{\ell^2} \zeta(2\sigma)^{1/2}$. At $\sigma \leq 1/2$, use the finite coefficient vector proportional to $(n^{-\bar{s}})_{n \leq N}$. Its unit-norm evaluation is $(\sum_{n \leq N} n^{-2\sigma})^{1/2} \rightarrow \infty$. Thus evaluation has no bounded extension at or below that boundary. This direct proof also specifies what “evaluation” means where not every infinite series converges [10].

Write $n = \prod_p p^{v_p(n)}$ and $\chi_n(z) = \prod_p z_p^{v_p(n)}$ on the product torus $\mathbb{T}^{\mathcal{P}}$. With product normalized Haar measure,

$$\int \overline{\chi_n} \chi_m dm_H = \delta_{nm}, \quad \lim_{T \rightarrow \infty} T^{-1} \int_0^T n^{it} m^{-it} dt = \delta_{nm}.$$

The first identity follows by integrating the finite coordinate product: unique factorization makes at least one nonzero exponent when $n \neq m$, and its circle integral is zero. The second follows from the actual Gram integral. A finite integer set may generate a lower-rank exponent lattice (for example 2, 4); this relation is retained by the torus construction. The half-density and evaluation boundary are two different analytic results, not a common arithmetic or physical theorem.

9 Induced metric, exact refinement and zero capacity

On the corrected D_{reg} take $g = S^* \delta_{\mathbb{R}^3}$ in its completed atlas, $\mathcal{H}_g = L^2(dA_g)$, and

$$\mathcal{E}(f, h) = \int g^{ab} \overline{\partial_a f} \partial_b h \, dA_g.$$

For nested finite-energy conforming spaces whose bases satisfy

$$\phi_i^{(n)} = \sum_{\alpha} J_{\alpha i} \phi_{\alpha}^{(n+1)},$$

sesquilinearity gives $J^* G_{n+1} J = G_n$ and $J^* K_{n+1} J = K_n$ exactly. These identities require exact integration; they assert no discretization convergence or solver accuracy.

Theorem 9.1 (Capacity and form removability).

Every one of the five rank-loss labels has local 1-capacity zero, using a full, half-, or quarter-neighbourhood as appropriate. Deleting $P_{\pm}$ does not change the closed form with fixed outer Dirichlet boundary.

Proof. For the standard cross-cap,

$$g = \begin{pmatrix} 1 + y^2 & xy \\ xy & x^2 + 4y^2 \end{pmatrix}, \quad w^2 = \det g = x^2 + 4y^2 + 4y^4, \quad \rho^2 = x^2(1 + y^2) + y^4.$$

Since ρ is ambient distance, $|\nabla_g \rho| \leq 1$ on the regular locus. For $0 < r < 1$, the set $\rho < r$ lies in $|x| < r$, $|y| < \sqrt{r}$ and $w \leq 3\sqrt{r}$, so its area $A(r) \leq 12r^2$. Choose a cutoff equal to one below ε^2 , zero above ε , and $\log(\varepsilon/\rho)/\log(1/\varepsilon)$ between. Smooth its corners on comparable annuli. Its squared L^2 norm is $O(\varepsilon^2)$, and its energy is bounded by

$$\frac{C}{\log^2(1/\varepsilon)} \int_{\varepsilon^2}^{\varepsilon} r^{-2} dA(r) \leq \frac{C'}{\log(1/\varepsilon)} \rightarrow 0.$$

The estimate follows from Stieltjes integration by parts: the endpoint contribution is bounded and $2 \int r^{-3} A(r) dr = O(\log(1/\varepsilon))$. These cutoffs are smooth source functions because they are constant near the singularity. Restricting them to half- or quarter-domains decreases their norms and energies.

The actual cross-cap equivalences in Proposition 2.1 compare the induced quadratic forms by positive constants: the target derivative and its inverse are bounded. In dimension two this also compares area densities and Dirichlet energies. Thus capacity transfers to all five actual germs; no metric isometry is required. Multiplying a compactly supported smooth test function by $1 - \chi_{\varepsilon}$ near $P_{\pm}$ approximates it in $\mathcal{E} + L^2$ norm, by the product rule and the displayed bounds. The reverse closure inclusion is immediate. This is equality of form closures, not of operator graph closures. $\square$

Theorem 9.2 (Actual compactness and positive Dirichlet gap).

The fixed-outer-Dirichlet form domain embeds compactly into $\mathcal{H}_g$. Its Friedrichs operator H_F has compact resolvent and satisfies $H_F \geq \lambda_D I$ for some $\lambda_D > 0$.

Proof. Use the original completed-source coordinates at each of the five points, translated to the origin, and write $R = (x^2 + y^2)^{1/2}$. The standard cross-cap has area density comparable to R . The actual source and target diffeomorphisms from Proposition 2.1, with bounded derivatives and inverses, therefore give $dA_g \asymp R dx dy$ in these coordinates. Also $g \leq CI$ since the actual extended map has bounded derivative. Consequently the energy dominates $c \int R |\nabla f|^2 dx dy$. The physical coordinate domains are a disk, half-disk or quarter-disk; their radial angular intervals are fixed. No metric cone equivalence is used.

Let χ be a chart cutoff, equal to one near zero and vanishing near its outer circular edge. For a smooth core function set $h = \chi f$. On each radial ray, integration of $(R|h|^2)'$ and Cauchy–Schwarz give

$$\int_0^{R_0} |h|^2 dR = -2\Re \int_0^{R_0} R \bar{h} \partial_R h dR \leq 2 \left(\int |h|^2 dR \right)^{1/2} \left(\int R^2 |\partial_R h|^2 dR \right)^{1/2}.$$

The endpoint terms vanish; the initial core avoids the singularity. Squaring and integrating the fixed angular interval yields

$$\int \frac{|\chi f|^2}{R^2} dA_g \leq C \int R |\nabla(\chi f)|^2 dx dy \leq C'(\mathcal{E}(f, f) + \|f\|^2).$$

The cutoff derivative lies on a regular compact annulus, where the last bound follows from the product rule. The inequality extends by form closure. In particular, the L^2 mass below $R < \varepsilon$ is bounded by $C\varepsilon^2$ times the form norm squared, uniformly at all five points.

Away from these disks the actual metric is smooth and uniformly elliptic on a compact source with ordinary Lipschitz boundary and retained regular corners. The usual Rellich theorem [9] gives compactness there. A diagonal subsequence followed by the uniform small-disk mass bound gives compactness on the whole source, including both endpoint corners.

If no positive gap existed, take $\|f_n\| = 1$ and $\mathcal{E}(f_n, f_n) \rightarrow 0$. Compactness gives a strong L^2 limit of norm one. On every regular chart its weak derivative is zero, hence it is constant on the connected interior. On a regular outer-boundary collar, the zero Dirichlet trace passes to the weak H^1 limit and makes this constant zero, a contradiction. This proves the gap. The form representation theorem and compact form embedding give the claimed compact resolvent. $\square$

For the standard metric an additional exact identity is

$$1 - |\nabla_g \rho|^2 = \frac{x^2 y^4}{(x^2 + 4y^2 + 4y^4)(x^2(1 + y^2) + y^4)} \leq \rho/4.$$

The front coordinates $x = r \cos \theta / \sqrt{1 + r \sin \theta}$, $y = \epsilon \sqrt{r \sin \theta}$, $0 < \theta < \pi$, have $\rho = r$. Each front has limiting length πr . Indeed its normalized speed converges to one away from the endpoints and is bounded by $C(1 + \sqrt{r/\sin \theta})$, with an integrable dominating function. The two fronts join on the source to one circle, so $L(r)/r \rightarrow 2\pi$. Wirtinger’s inequality on that circle gives

$$\int_{\rho < r_0} \frac{|u - \Pi_\rho u|^2}{\rho^2} dA_g \leq C\mathcal{E}(u, u)$$

for the standard model: coarea uses $|\nabla \rho|$ bounded above and below, and the tangential part of the energy controls the link derivative. In seam coordinates $x = \xi, y = \xi z$ the determinant is $\xi^4(1 + 4z^2 + 4\xi^2 z^4)$, and the density-weighted inverse metric has Schur complement $\sqrt{1 + 4z^2 + 4\xi^2 z^4}/[\xi^2(1 + 4z^2)]$. On fixed strips $|z| \leq K$ it bounds the weighted one-dimensional Poincare estimate. The transverse zero solutions, a constant and a primitive of $\sqrt{1 + 4z^2}$, are not localized in $L^2(\mathbb{R}, \sqrt{1 + 4z^2} dz)$. These calculations establish model inequalities. They alone do not exclude all defect modes of an actual patched unbounded operator.

10 The operator domain and its boundary obstruction

Let H_F be the self-adjoint operator associated with the closed fixed-outer-Dirichlet form. This definition uses form closure and makes sense before a point-trace theorem is available. Historical v12 instead set the punctured minimum equal to the graph closure of $-\Delta_g$ on $C_c^\infty(D^\circ \setminus \{P_+, P_-\})$. Denote that closure by H_0 .

Theorem 10.1 (Outer-boundary deficiency obstruction).

For that definition of H_0 , both deficiency spaces are infinite-dimensional. In particular, its claimed indices $(2, 2)$ and exhaustive $U(2)$ extension family are false. Fixing the outer boundary in a sentence does not alter the adjoint of the specified compact-core operator.

Proof. Choose a regular open outer-boundary arc away from all five singular labels and corners. It carries a smooth uniformly elliptic metric and the usual Sobolev trace. For any smooth boundary value h supported inside this arc, choose a smooth collar extension w_h , supported away from the singular points. For $z \in \mathbb{C} \setminus \mathbb{R}$ put

$$v_h = w_h - (H_F - z)^{-1}(-\Delta_g - z)w_h.$$

This is an L^2 function. Distributionally on the interior it satisfies $(-\Delta_g - z)v_h = 0$, so testing against the punctured compact core puts v_h in $\ker(H_0^* - z)$. The resolvent term belongs to the fixed Dirichlet form domain and has zero trace on the regular arc. Hence v_h has trace h . Linearly independent choices of h give linearly independent v_h . The argument works for $z = i$ and $z = -i$ and requires no claim about the cross-cap Green asymptotics. The Hilbert space is separable, so these deficiency dimensions are countably infinite.

An elementary interval witness illustrates the graph-closure distinction: $v(x) = x(1 - x)$ has Dirichlet values zero but $\langle -v'', 1 \rangle - \langle v, -1'' \rangle = 2$. The same pairing is zero for every interior compact test function and remains zero under graph limits. Thus even the ordinary Dirichlet domain is larger than that compact graph closure. $\square$

Theorem 10.2 (Conditional point restriction with fixed outer boundary).

Assume that the actual H_F domain admits a graph-continuous surjective trace $\tau : \text{Dom } H_F \rightarrow \mathbb{C}^2$ at the interior singularities, and that $\ker \tau$ is dense in $\mathcal{H}_g$. Then $A = H_F|_{\ker \tau}$ is closed, densely defined and symmetric, with deficiency indices $(2, 2)$. Its self-adjoint extensions are parametrized by $U(2)$. These hypotheses do not assert the actual existence of the traces.

Proof. The kernel is closed in the graph norm. For nonreal z , the bounded surjection $\tau(H_F - z)^{-1} : \mathcal{H}_g \rightarrow \mathbb{C}^2$ has kernel $\text{ran}(A - z)$. The adjoint of this map supplies two independent defect vectors, because $\ker(A^* - \bar{z}) = \text{ran}(A - z)^\perp$. Thus both dimensions are two. Von Neumann's unitary classification now applies to this corrected operator. The regular Dirichlet point-interaction construction [11, Proposition 2.1 and Theorem 2.2] uses exactly this restriction, but its interior elliptic H^2 point estimate is not a theorem for our degenerate Whitney metric. The proof here is an abstract conditional theorem. $\square$

The standard metric has the exact residual

$$\Delta_g \log \rho = \frac{2x^2y^2(3x^2y^4 + 5x^2y^2 + x^2 + 7y^6 + 5y^4)}{(x^2 + 4y^2 + 4y^4)^2(x^2(1 + y^2) + y^4)^2}.$$

To check integrability directly, set $R = (x^2 + y^2)^{1/2} < 1$. The numerator is at most $Cx^2y^2(x^2 + y^4)$, whereas the denominator is at least $(x^2 + y^2)^2(x^2 + y^4)^2$. Thus the residual is bounded by C/R^2 and $dA_g \leq CR dx dy$. Its p th power is bounded in integral by $C \int_0^{R_0} R^{2-2p} dR$, finite for $1 < p < 3/2$. On $\rho = r$, the preceding radial defect bound and link length limit give $r^{-1} \int_{\rho=r} |\nabla_g \rho| d\ell \rightarrow 2\pi$. This standard flux and residual motivate a logarithmic Green candidate. It does not prove that applying an actual resolvent to the remainder gives a continuous correction,

a graph-bounded point trace, or an exhaustive maximal-domain expansion. Those are different estimates. The required completion is a two-sided parametrix in specified front/seam weighted spaces, with an overlap estimate controlling the localized graph norm and proving the actual resolvent maps that L^p remainder into continuous functions. A global parametrix must also cover both endpoint corner charts, although compactness is already established in Theorem 9.2. The conifold assumptions in [12] do not hold automatically in the degenerating seam coordinates.

Proposition 10.3 (Reference-scale invariant boundary planes).

On $\mathbb{C}_a^2 \oplus \mathbb{C}_b^2$ with form $\omega((a, b), (c, d)) = 2\pi(\langle a, d \rangle - \langle b, c \rangle)$, the only complex Lagrangian plane invariant under every $R_t(a, b) = (a, b + ta)$ is $\{a = 0\}$.

Proof. For (a, b) in an invariant plane, subtract its R_1 image to see that $(0, a)$ belongs to the plane. Isotropy gives $2\pi|a|^2 = 0$, so $a = 0$. Dimension equality gives the stated plane; direct substitution checks invariance. This is a finite-plane theorem. Identifying it with the Friedrichs extension requires the actual Green coefficients and form/domain comparison, beyond Theorem 10.2. $\square$

Zero puncture capacity proves the fixed-boundary form removability of Theorem 9.1. A claim of uniqueness among all submarkovian extensions additionally needs a theorem classifying those extensions by the relevant Dirichlet forms with the same outer boundary. The old minimum admits other outer-boundary conditions, so that unrestricted uniqueness claim also fails. Concretely, close the same energy on smooth completed-source functions allowed to have nonzero outer trace. Such functions have finite local energy: bounded derivatives and the cross-cap metric give an integrable $O(1/R)$ energy density in source coordinates. The form is closable, as can be checked first on regular compact subdomains and then in the global weighted gradient space. Smooth contractions and the chain rule give the Markov property after closure. The associated Neumann form operator extends the interior compact operator and contains the constant function in its zero eigenspace. In contrast, the positive Dirichlet gap excludes that eigenfunction for H_F . This supplies two distinct submarkovian extensions of the old H_0 . Their existence is not altered by the finite-plane calculation.

Theorem 10.4 (Actual Krein and buckling results for the compact-core minimum).

The actual closed compact-core operator H_0 is densely defined, symmetric and strictly positive. Its Friedrichs extension is H_F . Its Krein extension satisfies

$$\text{Dom}(H_0)_K = \text{Dom } H_0 \dot{+} \ker H_0^*, \quad \ker(H_0)_K = \ker H_0^*.$$

The kernel is infinite-dimensional. The reduced positive spectrum is discrete with finite multiplicities, and its eigenvalues are the weak buckling eigenvalues on $\text{Dom } H_0$:

$$\langle H_0 v, H_0 \varphi \rangle = \lambda \langle v, H_0 \varphi \rangle \quad (\varphi \in \text{Dom } H_0).$$

Proof. Interior compact smooth functions are dense in $\mathcal{H}_g$ by truncation and smooth approximation on the regular locus. Integration by parts makes the initial operator symmetric, and its graph closure is contained in H_F . The positive gap in Theorem 9.2 passes to this closure. The capacity theorem identifies the two Friedrichs forms, hence its Friedrichs extension is H_F . For the kernel repeat the collar construction in Theorem 10.1 at $z = 0$; the bounded inverse H_F^{-1} now exists. Independent boundary values inject into $\ker H_0^*$. The abstract Krein/buckling theorems apply to this actual closed densely defined strictly positive operator [14, Hypothesis 2.2, Theorem 2.4 and Theorem 3.4]. Compact resolvent of H_F gives discreteness of the reduced positive Krein spectrum. This retains the valid global spectral conclusion while correcting the false finite-channel dimension. $\square$

Under the point-trace hypotheses of Theorem 10.2, the corrected A is also strictly positive by restriction, and the abstract Krein theorem gives

$$\text{Dom } A_K = \text{Dom } A \dot{+} \ker A^*, \quad \ker A_K = \ker A^*.$$

If actual logarithmic boundary coordinates and zero-energy Green vectors are established, this domain becomes $b = R_0 a$, with R_0 Hermitian and reference transformation $R_0 \mapsto R_0 + tI$. Its reduced nonzero spectrum obeys the weak buckling equation $\langle Av, A\varphi \rangle = \lambda \langle v, A\varphi \rangle$ on $\text{Dom } A$ [13, 14]. Its Friedrichs form must be compared with the fixed Dirichlet form before transferring that compactness conclusion to A . The actual point traces, finite trace rank and logarithmic coordinates remain unproved. The global compactness result above concerns H_F and the explicitly defined H_0 ; it does not supply these missing Green coefficients.

11 Normalization defect and the support obstruction

Let X be a sufficiently small standard cross-cap image, and $\nu : \tilde{X} \rightarrow X$ its proper topological normalization, obtained from the source map. It is one-to-one off the open double ray L and two-to-one on L , with one preimage at the pinch. In the ordinary sheaf category,

$$\mathcal{Q}_\nu = \nu_* \mathbb{Q}_{\tilde{X}} / \mathbb{Q}_X \simeq j! \mathcal{L}_{\text{sgn}}, \quad j : L \hookrightarrow X.$$

Here j is locally closed; extension by zero means the open-ray extension inside its closed support, followed by the closed direct image. Indeed the diagonal stalk map is an isomorphism at one-preimage points and has one-dimensional anti-invariant quotient at two-preimage points. Properness gives these direct-image stalks by shrinking neighbourhoods. The quotient pinch stalk is zero. The restriction is the sheet-sign line; the extension-by-zero description follows by the adjunction map and its stalk isomorphisms [15].

Let i include the pinch. Every ordinary morphism $j! \mathcal{L}_{\text{sgn}} \rightarrow i_* \mathbb{C}$ is zero, since every stalk map is zero. For the local ray the nearby group $H^0(i^* Rj_* \mathcal{L}_{\text{sgn}})$ and degree-one costalk $H^1(i^! j! \mathcal{L}_{\text{sgn}})$ are one-dimensional. One may compute the latter from the compactly supported cohomology of $(0, \varepsilon)$, or the local recollement triangle. Sheet exchange acts by -1 . A linear equivariant map from that line to a sheet-even line must satisfy $T(-v) = T(v)$ and therefore vanishes in characteristic zero. This sheaf theorem is independent of the existence of an actual scalar Green line. The standard logarithm is sheet-even because $\rho(0, y) = \rho(0, -y)$; exporting this interpretation to an actual exhaustive Green space still depends on the analytic construction above.

An underlying compact or condensed set does not fix a structure sheaf: even a one-point space supports both $\mathbb{Q}$ and $\mathbb{Q}[\epsilon]/(\epsilon^2)$. The chosen algebraic fiber instead has its explicitly supplied coordinate ring $\mathbb{Q}[X, Y]/(X^2 + Y^4 - 1)$. Neither its structure sheaf nor its mixed Hodge data is recovered from a bare topological normalization defect.

12 Phase lines, cone models and transmission hypotheses

The squaring cover pulled back by $F/|F|$ supplies a flat Hermitian sign line on each punctured interior source disk. Theorem 3.1 gives its holonomy -1 , with no continuous holonomy parameter. On an exact round cone the scalar twisted angular eigenvalues are $n + 1/2$, and the radial indicial powers are $r^{\pm|n+1/2|}$. Since $r^\lambda \in L^2(r dr)$ exactly for $\lambda > -1$, the two weights $n = 0, -1$ are limit-circle channels. This scalar model has two lowest channels, not a preferred one-dimensional normalization defect.

For the disk-bounding spin structure the untwisted angular weights are also $\mathbb{Z} + 1/2$. Twisting by the sign line makes them $\mathbb{Z}$. The chiral exact-cone operator has the form $c(dr)(\partial_r + 1/(2r) + A_\partial/r)$. On its angular zero eigenspace, $r^{-1/2}f(r)$ gives a unitary map from $L^2(dr)$, reducing

the operator to $c(dr)\partial_r$. Locally at the tip, H^1 has an endpoint trace and the minimal closure has zero trace. Thus that exact block contributes one complex endpoint coefficient per chirality. The full block contributes two. This is a local tip quotient after excluding the regular outer-boundary trace, not the unrestricted maximal/minimal quotient on a bounded punctured disk. The latter contains infinitely many regular boundary traces.

The intrinsic periodic Dirac spectrum on any metric link circle of length $L(r)$ is $2\pi\mathbb{Z}/L(r)$ once its total flat holonomy is trivial. This link result does not construct a parametrix for the surface Dirac operator. The model inverse for $\partial_r + k/r$ is

$$Q_k f(r) = \begin{cases} r^{-k} \int_0^r s^k f(s) ds, & k > 0, \\ -r^{-k} \int_r^{r_0} s^k f(s) ds, & k < 0, \\ \int_0^r f(s) ds, & k = 0. \end{cases}$$

Differentiation checks $(\partial_r + k/r)Q_k f = f$. To patch such inverses, one must prove boundedness in chosen graph spaces and control the overlap remainders. The claim that all cutoff commutators gain positive weight is false without further construction: under $r = \varepsilon R$, $\varepsilon \partial_r \chi(r/\varepsilon) = \chi'(R)$. Composing that commutator with a first-order model inverse leaves weight zero, not a factor tending to zero. Shrinking alone therefore does not justify the claimed Neumann series. Cone-operator theory [16, 17] cannot be imported until the actual Whitney seam and domain hypotheses are verified. The actual chiral/full trace dimensions in v12 are consequently unproved.

Proposition 12.1 (First-order phase/deck comparison).

The pullback $h_F(v, w) = \Re(dF(v)dF(w))$ is positive definite at $P_\pm$. Its restriction normalizes the kernel of dS . For $k_\pm = \partial_t + c_0 s_\pm \partial_u / [2(1 - c_0)]$,

$$dS(k_\pm) = 0, \quad |dF(k_\pm)|^2 = 655/4 - 72\sqrt{5} > 0.$$

The two normalization branches have opposite first-order F phases. Their four limiting square roots form a nonsplit $\mathbb{Z}_4$ torsor over the two branches. A chosen orthogonal source reflection lifts to Pin and exchanges the two chiral Clifford lines.

Proof. Invertibility of dF proves positive definiteness. Actual differentiation gives $dF(k_\pm) = s_\pm(1 - 2c_0) - is_\pm(15c_0 - 4)/[2Q_0(1 - c_0)]$; reduction by $c_0^2 = 3c_0 - 1$ gives the squared norm. The two double branches approach in directions $\pm k$, so their normalized phases tend to $a, -a$. If $\alpha^2 = a$, their roots are $\{\alpha, i\alpha, -\alpha, -i\alpha\}$. Multiplication by i squares to the phase-deck sign, and the extension $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2$ has no group section. Finally Clifford multiplication by a unit reflecting vector anticommutes with chirality, and its two Pin lifts differ by a central sign [18]. $\square$

The reflection requires a chosen positive metric and either a Pin^+ or Pin^- convention. It is not an automatic isometry of the actual induced metric. Right-left cross-cap equivalence only compares metrics: for example the target shear $(X, Y, Z) \mapsto (X, Y, Z + Y)$ preserves the cross-cap germ but destroys the standard $y \mapsto -y$ source metric symmetry. Likewise a Pin map of the spin fibers does not supply an identification of the separate phase-line fibers. Their first-order quarter-turn is precisely the distinction that the four-phase torsor records.

Proposition 12.2 (Conditional full-spin cut action).

*On regular paired cut banks assume specified trace spaces, Dirac Green's formula, and a unitary full-spin bundle map U including a phase-line identification, such that $U^*c(\nu_2)U = -c(\nu_1)$. Then the graph $(v, \lambda Uv)$ is isotropic for the Green form exactly when $|\lambda| = 1$, and is fiberwise maximal isotropic. The real Green transgression coefficient is uniquely $1/2$. If a fixed Pin reduction is additionally required, scalar multiples of its chosen lift remain Pin lifts exactly for $\lambda = \pm 1$.*

Proof. The Green pairing restricted to the graph is $(1 - |\lambda|^2) \int \langle c(\nu_1)v, w \rangle ds$. The boundary Clifford map is invertible, so isotropy is equivalent to the coefficient being zero. The graph has half the doubled fiber dimension, giving fiberwise maximality. For $x = \sum_j \langle \psi_j, D_j \psi_j \rangle$, the self-pairing Green term is $\bar{x} - x$. Hence $x + a(\bar{x} - x)$ is real for arbitrary x exactly when $a = 1/2$. Its variation has no boundary term on a common isotropic trace space. This does not single out this graph among all isotropic conditions. The kernel of the chosen Pin double cover is $\{1, -1\}$, giving the last claim once that reduction has been supplied [17, 18]. $\square$

The conditional trace theorem neither identifies the phase and normalization lines nor fixes their relative unitary trivialization. Without that additional input a continuous phase freedom remains. It also leaves an overall action multiplier and a clock unspecified. End behaviour of the singular cut must be included before turning the regular-bank calculation into a global self-adjoint Dirac realization.

For the self-adjoint circle operator $-i\partial_\theta$ with H^1 boundary condition $f(2\pi) = -f(0)$, the eigenvalues are $n + 1/2$. The involution $n \mapsto -n - 1$ pairs each with its negative. Thus its eta series vanishes absolutely for $\Re s > 1$ and its meromorphic continuation has value zero at $s = 0$ [19]. This elementary model produces no net spectral chirality.

13 Structural limits and the remaining research problem

Several negative conclusions need little machinery but prevent unjustified identifications. A residual does not fix communication capacity: on identical binary labels a constant stochastic channel has capacity zero and a noiseless one has capacity $\log 2$. The kernel and resource constraints are absent from the residual. A proposed sphere-volume to disk-area ratio with $r = \phi^{-2}R$ is $2\pi\phi^4 R$ and scales as a length; dividing by R defines a dimensionless number but no mass operator. For the canonical Kahler form on $\mathbb{C}^2$, the power coordinates depend on disjoint canonical pairs and satisfy $\{|z_+|^2, |z_-|^2\} = 0$. The full $SU(2)$ moment map is a different observable algebra. A nonempty proper subset of $SU(2)$ cannot be invariant under its full transitive left action; this conclusion is conditional on an actual proper source lift, which (1) alone has not specified.

The chosen complex residual supplies the smooth algebraic pair (E, D) needed for its mixed Hodge calculation. Real signal topology and the number $\sigma = 1/2$ do not specify such a pair, a finite-field model, Frobenius, or an ℓ -adic sheaf. A single curve is not a Weil cohomology theory. For a separately specified realization extended to open varieties with localization and Tate twists, its localization sequence has boundary term the kernel of the sum map $H^0(D)(-1) \rightarrow H^2(E)$, of dimension one. That is a conditional evaluation of an existing theory.

The outstanding actual analytic problem is concrete. Construct the corrected fixed-boundary point traces and a two-sided Green/Dirac parametrix with uniform overlap graph estimates. Prove the needed resolvent regularity and exclude additional modes. Global compactness and the positive fixed-Dirichlet gap are established above, including both endpoint corners. Specify the spin domains, Pin convention and phase-line transmission data before claiming an actual trace interface. The earlier proof labels of v12 do not resolve these obligations; the domain counterexample shows why they must be revisited. The source geometry, capacity theorem and independent spectral results remain available while this analytic work continues.

A Claim audit, formal coverage and reproducibility

The immutable historical authority is the 43-page final v12 PDF with SHA-256

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Physical and printed page numbers agree; Section 7 is on pages 14–16, not 13–15. The full machine-readable and human-readable claim maps pin 66 named blocks, including definitions and remarks, to this PDF. Abstract, conclusion, figure and historical status-table assertions inherit those audited results, not their former “proven” labels.

The accompanying Lean project uses Lean/Mathlib 4.33.1 with the unchanged RuledV5 and StokesV5 source dependencies. It proves actual atlas/rank and small-circle statements through those identical maps, a genuine polar fold through existing smooth inverse charts, actual Mellin interval integrals, actual Gram entry integrals and eigenvalue estimates, and selected finite boundary identities. The separate companion lists precisely what is encoded. Capacity integrals, Hilbert operator domains, cohomology, sheaf theory and the actual Green/Dirac parametrices are not certified by that finite inventory. External Whitney recognition, Milnor comparison, Gysin/MHS and abstract Krein/buckling results remain cited dependencies with explicit hypotheses.

Independent exact replay rebuilds the actual derivative determinants and Euclidean jet invariants, the polar remainder correction, weighted adjoint coefficient, model logarithmic residual and counterexample calculations. It supplies neither an infinite-dimensional parametrix nor an unbounded operator-domain certificate. The two registered ancestor figures were reviewed: the monodromy schematic is useful but can be mistaken for a uniform clock; the golden-rectangle image is correct background but adds no proof. The new source diagram and comparison table instead guide the corrected statements. All ancestor files and historical sealed artifacts remain unchanged.

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